Patent valuation

What is a patent worth, and how can we tell?

Innovation Management · A.Y. 2026/27
Prof. Grazia Sveva Ascione · Wednesday 4 November 2026

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Today's lecture

01	**Why value a patent?**	when we need it, and what we know about patent value
02	**Financial valuation**	a value in euros: cost, market and income methods
03	**Three dimensions of patent quality**	economic, legal, technological
04	**Technological value indicators**	inventors, scope, citations, breakthroughs, generality, originality, novelty
05	**Legal value indicators**	grant, examination, opposition, claims and their text
06	**Economic value indicators**	renewals, family, market reaction, transfers, products
07	**Scoring patents**	rule-based algorithms, tools, the IP scoreboard

01

Why value a patent?

When valuation is needed, and what we know about patent value

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When do we need to know what a patent is worth?

Investing in R&D

Was the R&D money well spent?

Managing the portfolio

Renew, drop, sell or transfer a patent?

Buying or selling a firm

Mergers, acquisitions, joint ventures, bankruptcies

Going to court

Damages for infringement, settlements

Licensing or selling

Setting the royalty or the price

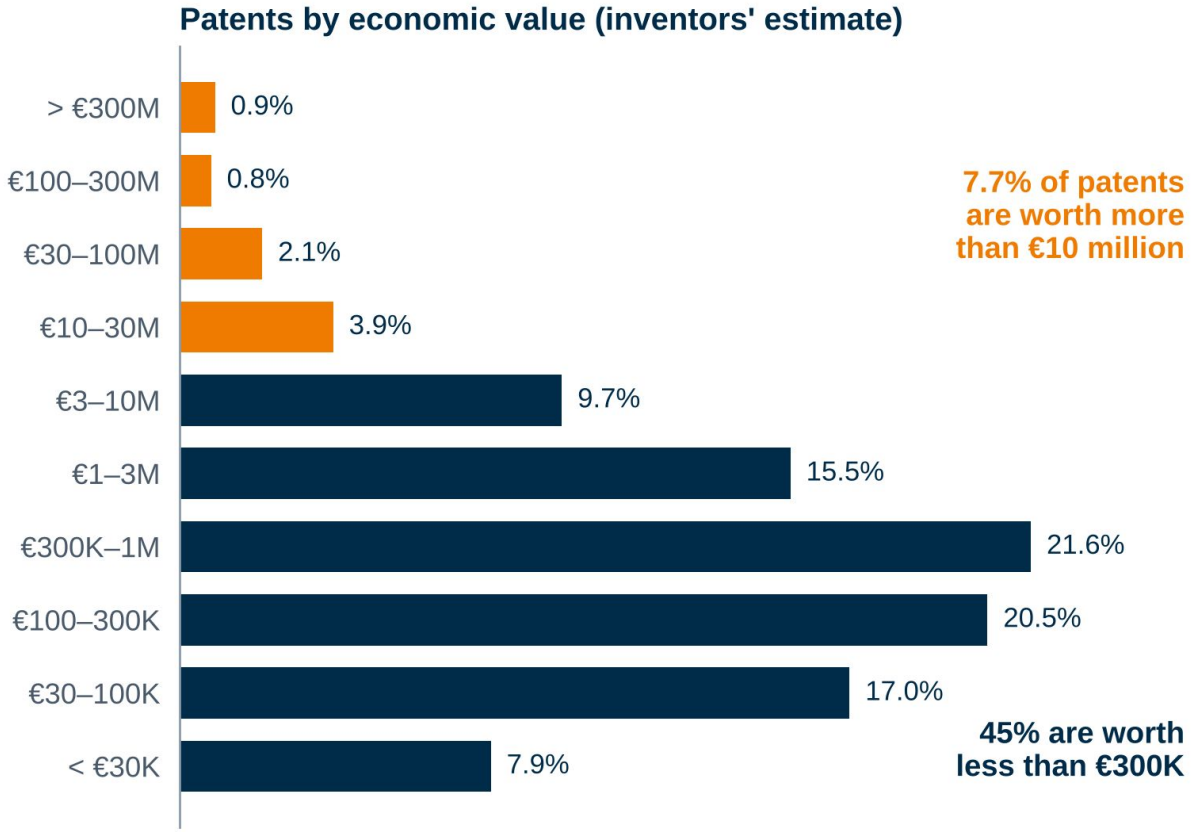
Raising money

Bank loans, venture capital

Accounting and taxes

Intangible assets in the balance sheet, tax regimes

Patent value is extremely skewed



A few winners, many patents of little value

Inventors of European patents were asked: what is the lowest price at which the owner would have sold the patent, on the day it was granted?

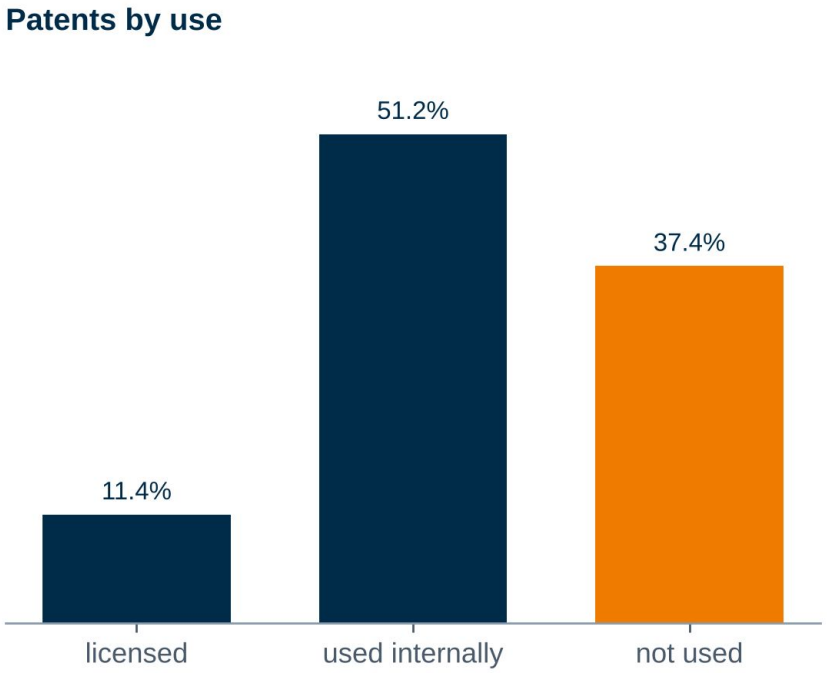
45% of patents are worth less than €300K; only **7.7%** more than €10 million.

So the average patent does not exist: the mean is pulled up by a handful of very valuable ones.

For a valuation this means great uncertainty: most patents in a portfolio will be worth little, and we rarely know in advance which ones will not.

Gambardella, A., Harhoff, D. & Verspagen, B. (2008), The value of European patents, European Management Review, 5(2), 69–84 (PatVal-EU survey).

Most patents are not used



37% of patents are not used: some block competitors, many are simply asleep.

51% are used inside the firm that owns them.

11% are licensed to others: the market for patents is thin.

So: many patents have little value and could not be sold or licensed anyway; others could be monetised, if someone could tell which.

Gambardella, A., Harhoff, D. & Verspagen, B. (2008), European Management Review.

Valuation is not pricing

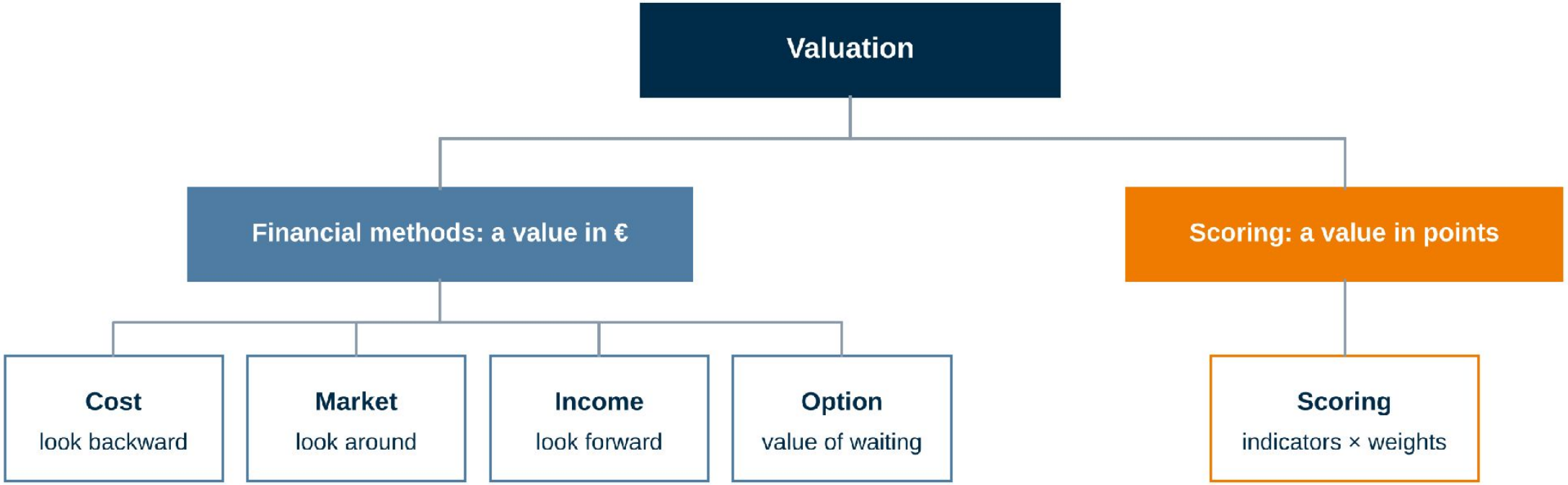
Valuation estimates **what a patent is worth**, mixing objective data and judgement. **Pricing** uses that estimate to reach an **agreement between buyer and seller**.

	Patent	Firm	Technology and market
Question	Is it strong and broad?	How does it fit the owner?	Is the market growing? Are there rivals?
Data	legal and bibliometric	business model, strategy, resources	market trends, innovation trends
Source	patent databases	internal documents	technology intelligence, market reports
Access	public	private	mixed
Cost	medium	low	high

Why it is hard: markets for patents are opaque and **the seller knows much more than the buyer** (*asymmetric information*).

Arora, Fosfuri & Gambardella (2001); de Rassenfosse et al. (2016).

A map of the methods



Why a patent can be valued on its own: it is identifiable, legally protected and enforceable, has a measurable economic life, and can be transferred.

The option method is beyond the scope of this lecture.

02

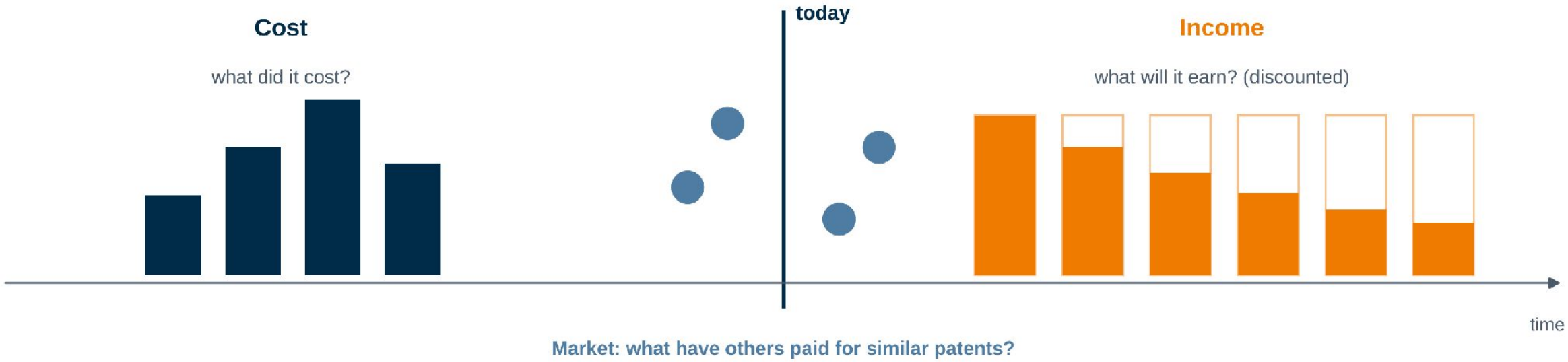
Financial valuation

A value in euros: cost, market and income methods



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Three ways to look at a patent



Cost: What would it cost to develop it again?

Market: What have others paid for similar patents?

Income: What income will it earn, in today's euros?

The cost method: what did it cost?

Two versions

Reproduction cost: the cost of making an exact copy of the asset.

Replacement cost: the cost of obtaining an equivalent asset, with the same use or function.

What to count: direct costs (salaries of researchers, materials, contractors), indirect costs (labs, equipment, administration) and opportunity costs (profits lost while developing it).

Historical cost can mislead

Too high: dead ends, inefficiencies, or a technology that is cheaper today.

Too low: inflation since the money was spent.

The main problem: value comes from future benefits, not from sunk costs. The method ignores how much the patent will earn, and for how long.

The market method: what have others paid?

The idea

Infer the value from the prices agreed in open-market deals for similar rights, as with the multiples used to value unlisted firms.

We need deals that are public (price and terms known), similar, and recent.

Why it is hard: patents are unique; sale prices are almost never published (licences more often); deals bundle patents with other assets.

Where to look for prices

- Royalty databases (RoyaltySource, RoyaltyStat, Knowledge Express)
- Journals on licensing (Licensing Economics Review)
- Court decisions on damages for infringement
- Annual reports of listed companies
- Patent auctions

Patent auctions: a market that did not take off

How it worked

Ocean Tomo pioneered live auctions of patents, like auctions of art.

An auctioneer takes the bids for each lot; the highest bid wins if it beats the seller's reserve price.

Ocean Tomo earned a 10% premium from the buyer and 15% from the seller.

Why it failed

Too few buyers and sellers to make a real market.

Adverse selection: buyers suspected that the patents on sale were the ones nobody wanted, of low or uncertain quality.

In 2009 Ocean Tomo sold its auction business to the broker ICAP, for $10 million.

Akerlof (1970) for the market for lemons.

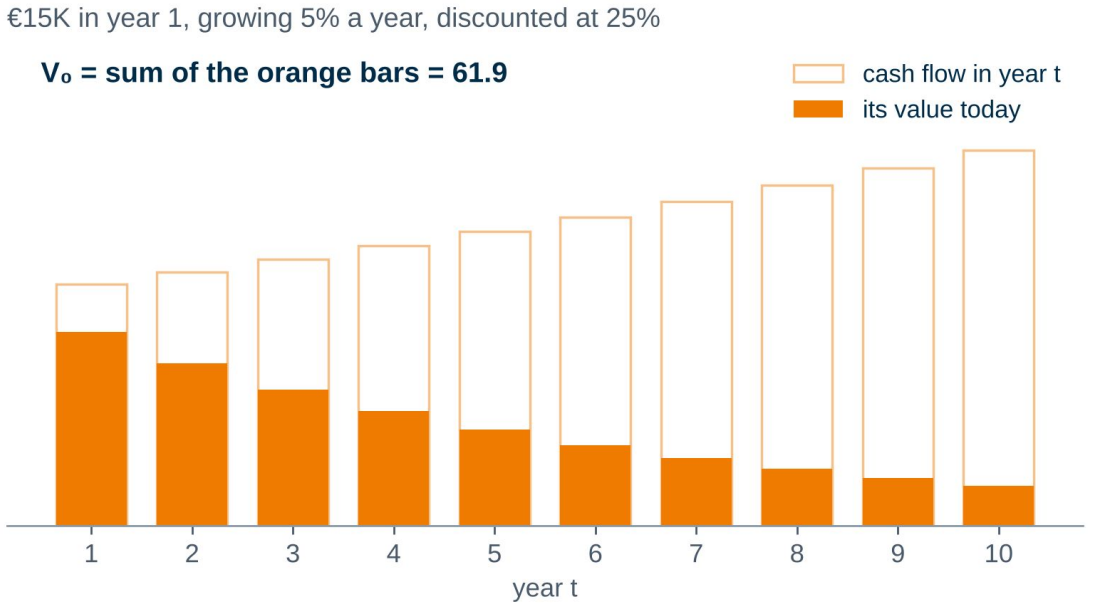
The income method: what will it earn?

$$V_0 = \sum_{t=1}^{T} \frac{CF_t}{(1+r)^t} + \frac{TV}{(1+r)^T}$$

CF cash flow the patent generates in year t, up to its end T

r discount rate: the higher the risk (market, technology), the higher r

TV terminal value: an advantage that may last after the patent expires



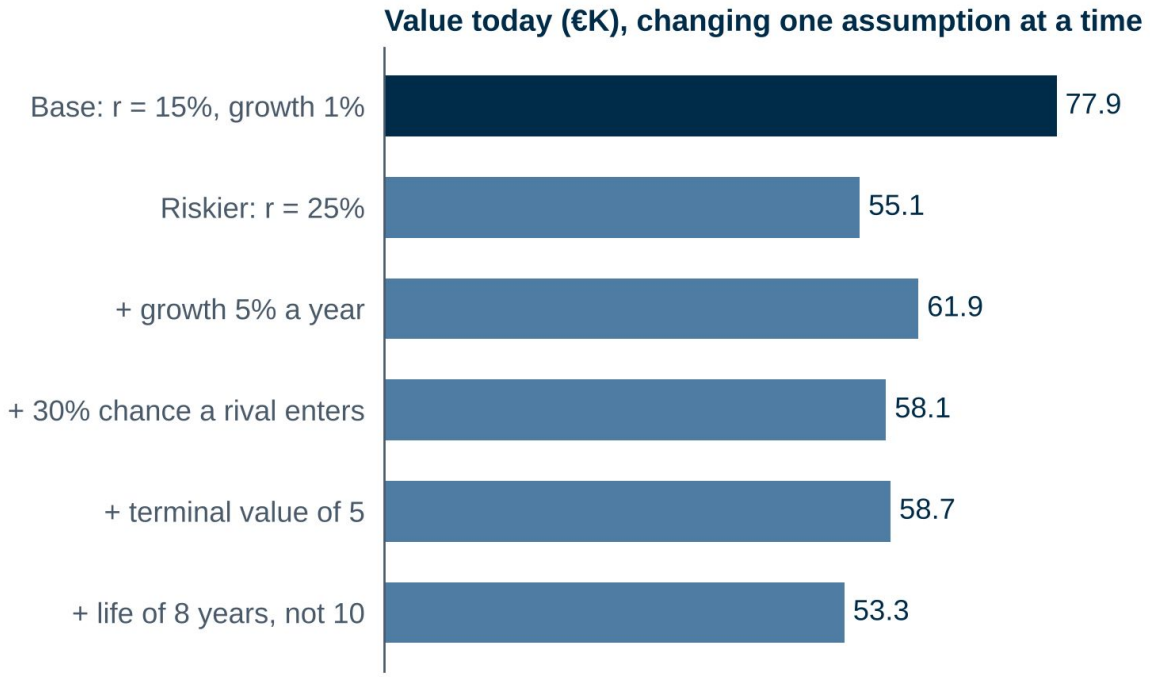
Two ways to get the cash flows

Discounted cash flows: from the business plan of whoever uses the patent.

Relief from royalty: the royalty you would pay if you did not own the patent (e.g. 3% of sales) is what you save.

Each orange bar is the value today of one year's cash flow: at 25% a year, €1 in year 10 is worth 11 cents today.

Small changes in the assumptions, large changes in the value



What moves the value

The discount rate. From 15% to 25% the value falls by almost a third. r must reflect market risk (demand, rival solutions) and technological risk (how mature the technology is, how much investment it still needs).

Growth, rivals, life. Each changes the value by 5–10%.

Cash flows. They must include everything the patent changes: revenues and costs, investments, taxes, working capital.

Illustrative example: €15K cash flow in year 1, 10-year life; a rival may enter after year 3 and cut cash flows by 40%.

The limits of the income method, in practitioners' words

50%

of IP managers find valuation essential to their strategy

30%

had not considered it, or found it unhelpful

> *"Valuations often focus on financial analysis which does not reflect the complexities of assets such as patents and trademarks."*

> *"For early-stage technology, the layers of uncertainty reduce the plausibility of income-based valuations."*

> *"For technology that is already in use, it is difficult to disaggregate the earnings of contributory assets."*

> *"Use of data from historic transactions to value intellectual property can be superficial and misleading."*

Global survey of IP brokers, IAM

Universities and start-ups: from euros to scores

1 · Not in use

Many patents are not yet part of a product: there are no cash flows to discount.

2 · Unknown markets

Little knowledge of the cost structure of the markets where the technology could go.

3 · Early stage

Low technology readiness levels: years of development before any income.

> So they start from **indicators** measured on the patent itself, and combine them into a score. But what makes a patent good?

03

Three dimensions of patent quality

Economic, legal, technological

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Value, quality, strength: three dimensions

The same patent can be judged by three stakeholders, who want different things from it.

Economic quality

Who cares: the owner, investors

Can it generate value, influence markets, open business opportunities?

It depends on market demand, rivals, licences, and on how much the owner invests in it.

Legal quality

Who cares: the owner, in court

Does it protect the invention, and will it hold if challenged?

It depends on patentability, validity, the scope of protection, infringement.

Technological quality

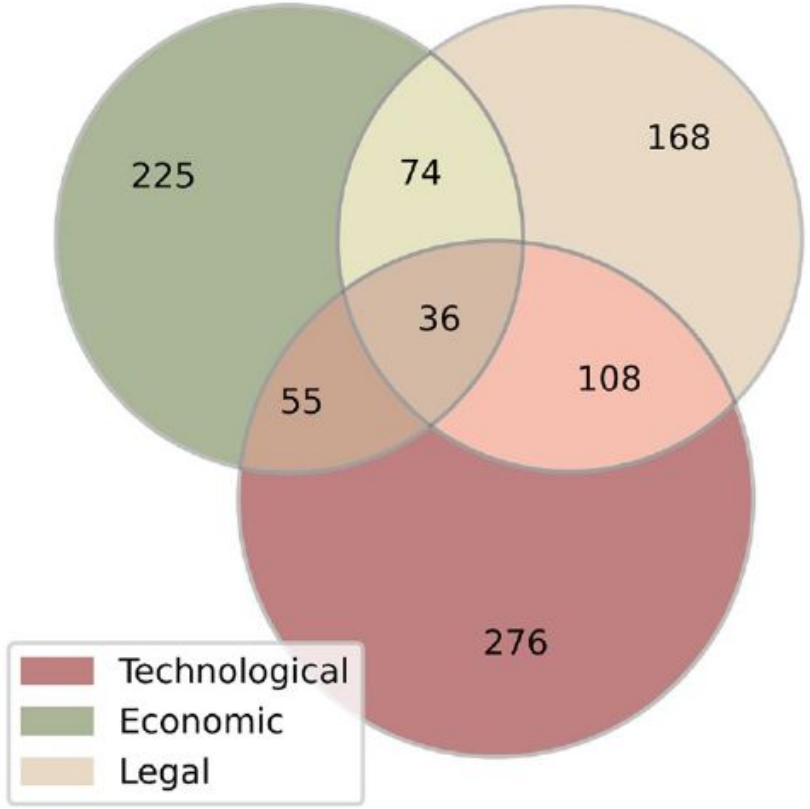
Who cares: the state and society

Is the solution new, and does it push technology forward?

It depends on originality, and on the contribution to later inventions.

Schmitt, V. J. (2025), Disentangling patent quality: using a large language model for a systematic literature review, Scientometrics, 130, 267–311.

What the literature measures: 985 indicators in 762 papers



Number of indicators in each dimension and overlap.

Most used indicators	Papers	Dimension
Forward citations	389	T
Family size	172	E · L · T
Number of claims	136	L · T
Backward citations	117	L · T
Renewals	87	E · L
Technology classes (IPC)	76	T
Inventors	66	T
Citations to science	55	L · T
Generality	33	E · T
Assignees	31	E · T

E economic · L legal · T technological

Two thirds of the studies measure technological quality; the economic one is the least measured, as its data are private.

Schmitt (2025), Scientometrics, Table 4 and Fig. 3.

The indicators of this lecture, by dimension

04 **Technological**	inventors · technological scope · backward citations · citations to science · forward citations · breakthrough inventions · generality · originality · text novelty
05 **Legal**	grant status · grant lag and examination · opposition and litigation · number of claims · length of the first claim · readability of the claims · abstractness of the claims
06 **Economic**	renewals · family size · co-applicants · market reaction to the grant · transfers and licences · products on the market

04

Technological value indicators

Inventors, scope, citations, generality, originality, novelty

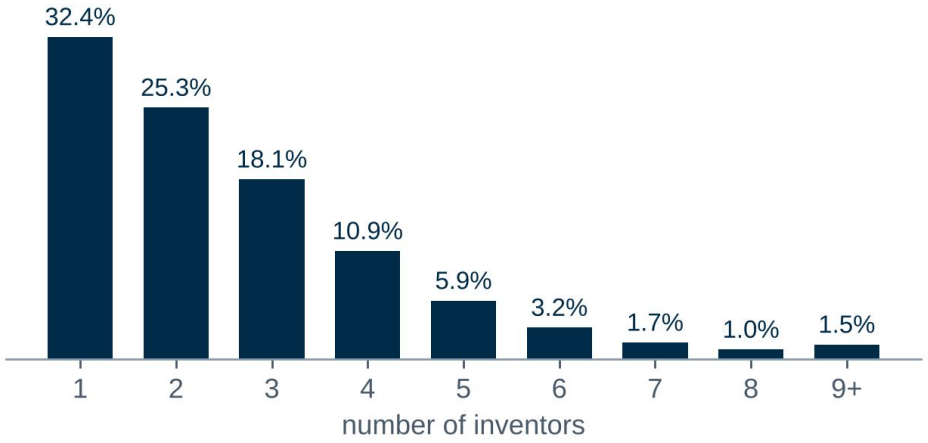
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Dimension **Technological**

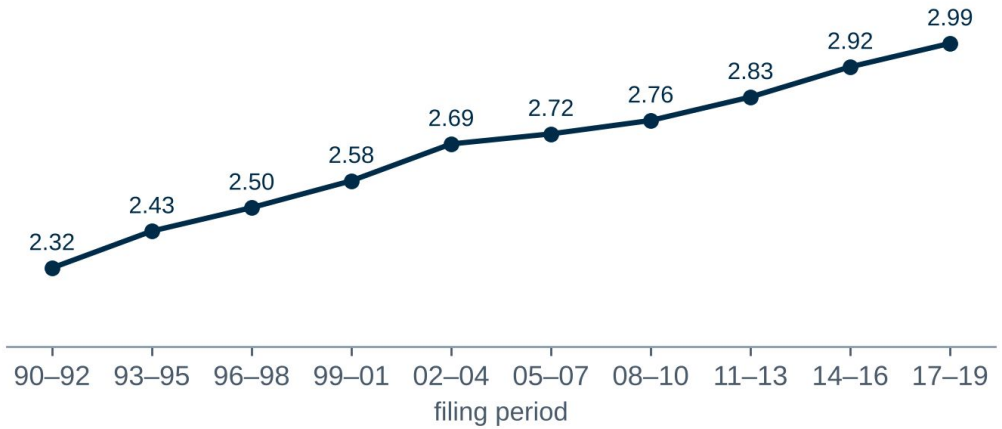
PATSTAT Data Catalog: TLS201_APPLN · TLS207_PERS_APPLN · TLS206_PERSON

Inventors: the research effort

Inventors per EP patent



Average team size grows



What it measures: the number of people who made the invention.

Why it signals value: firms put more people on the inventions they expect to be promising; bigger teams combine more knowledge.

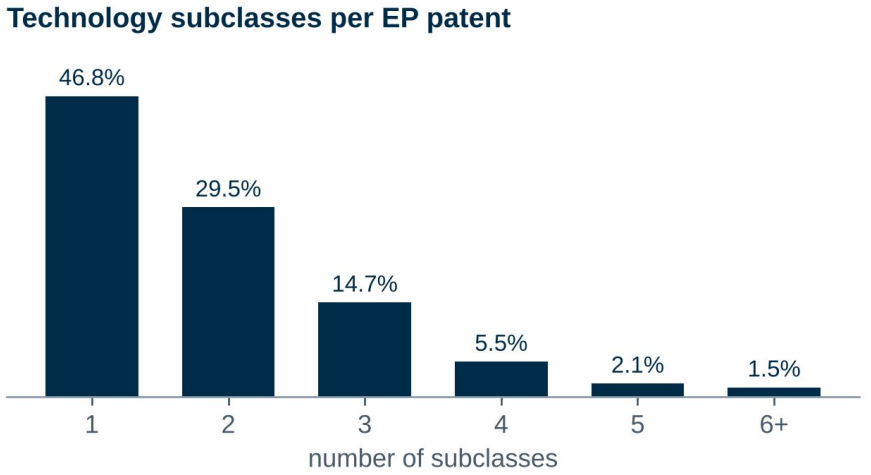
How to measure it: count the inventors of the application (TLS207 with invt_seq_nr > 0, or nb_inventors in TLS201).

Watch out: teams grow over time and differ by field, so compare patents of the same year and field. The same inventor can appear with different spellings.

Dimension **Technological**

PATSTAT Data Catalog: TLS209_APPLN_IPC · TLS224_APPLN_CPC

Technological scope: how many fields?



What it measures: the number of different technology fields in which the examiner classifies the invention.

Why it signals value: an invention that belongs to many fields can be used in more products and processes. Firms with broader patents have a higher market value (Lerner, 1994).

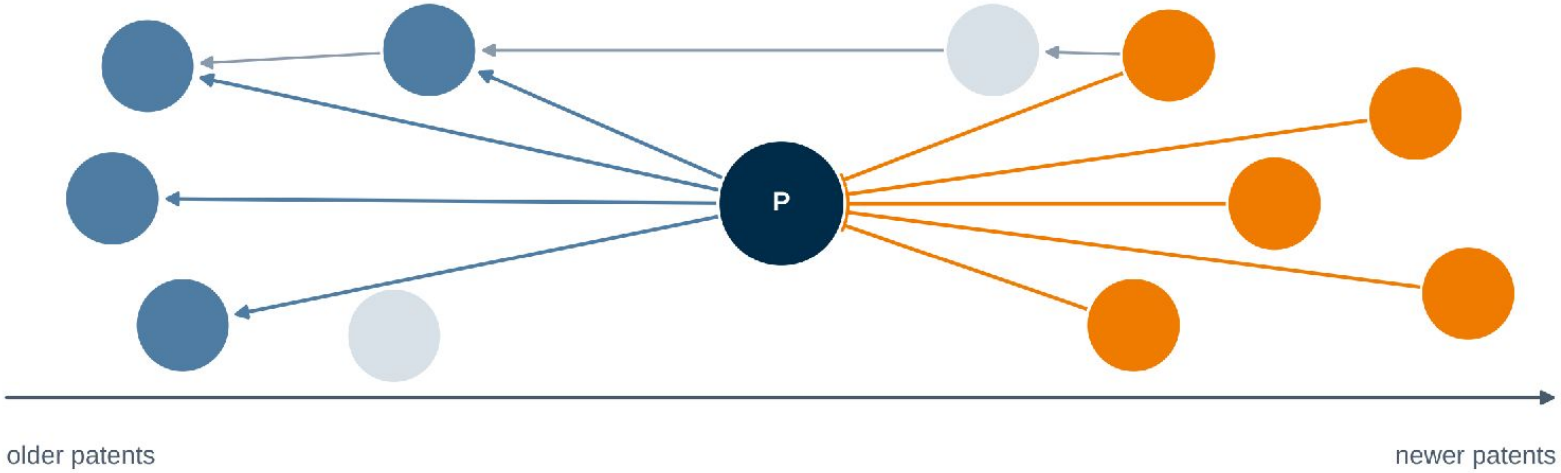
How to measure it: count the distinct **4-character CPC subclasses** of the application (TLS224), e.g. B65D and A47J.

OECD indicator · Patent scope: number of distinct 4-digit IPC subclasses of the patent

Dimension **Technological**

PATSTAT Data Catalog: TLS211_PAT_PUBLN · TLS212_CITATION · TLS215_CITN_CATEG

Backward citations: what it builds on



Forward citations of P

5 later patents cite P

Backward citations of P

P cites 4 earlier patents

arrows go from the citing to the cited patent

What it measures: how many earlier patents the patent cites (blue in the network).

Why it signals value: it builds on more earlier knowledge; and its claims survived a search that found a lot of close prior art.

How to measure it: TLS212 citations of all the publications of the application, brought back to the cited applications (as in Tutorial 3).

Watch out: most EP citations are added by the examiner; X and Y citations (TLS215) mean close prior art, which can also signal a weaker patent.

OECD indicator · Backward citations: number of patent documents cited (NPL excluded, self-citations included)

Dimension **Technological**

PATSTAT Data Catalog: TLS212_CITATION · TLS214_NPL_PUBLN

Citations to science: the link with research

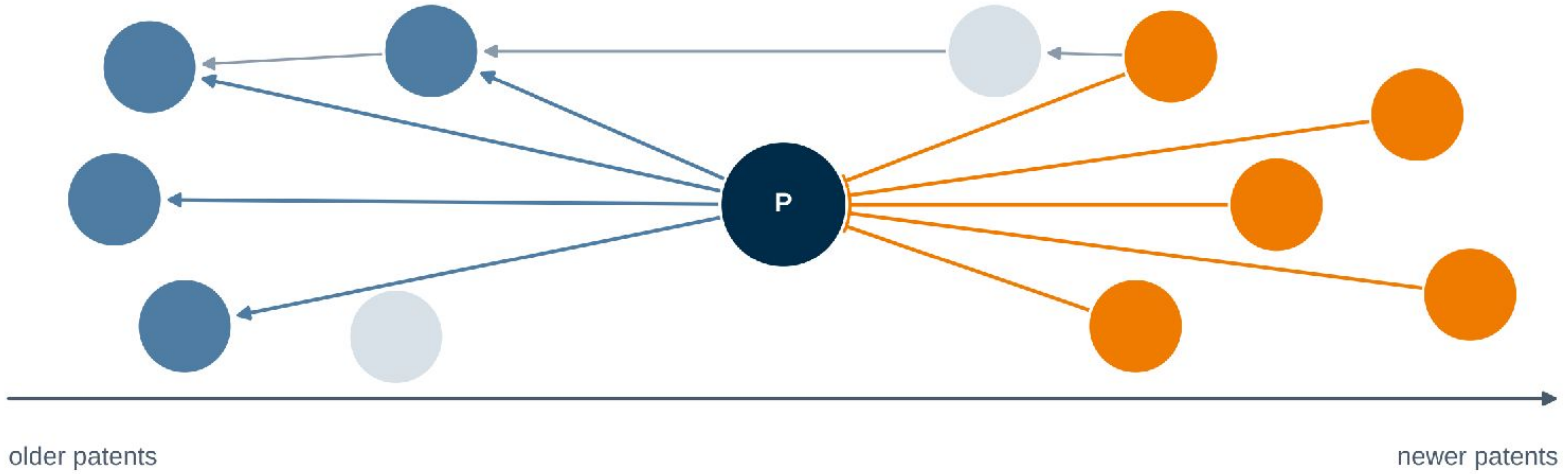
What it measures	How many scientific papers, books and other non-patent documents the patent cites.
Why it signals value	Inventions close to science are often more radical and more general.
How to measure it	TLS212 citations with cited_npl_publn_id ≠ '0', then TLS214 for the type (Tutorial 3); or *Reliance on Science*.
Watch out	Not all non-patent literature is science: catalogues, standards and web pages are cited too.

OECD indicator · NPL citations: number of non-patent documents cited

Dimension **Technological**

PATSTAT Data Catalog: TLS212_CITATION · TLS201_APPLN

Forward citations: the technological impact



Forward citations of P

5 later patents cite P

Backward citations of P

P cites 4 earlier patents

arrows go from the citing to the cited patent

What it measures: how many later patents cite this one (orange in the network). 81% of EP patents receive none from other EP patents.

Why it signals value: later inventions build on it, so it opened a road others followed. Cited patents are renewed longer and litigated more, and owners value them more (Harhoff et al., 1999).

How to measure it: count the citing families (TLS212 back to TLS211 and TLS201; or nb_citing_docdb_fam in TLS201).

Watch out: many citations are added by examiners, not by the inventors; citations from the same firm (self-citations) say more about the firm than about the field.

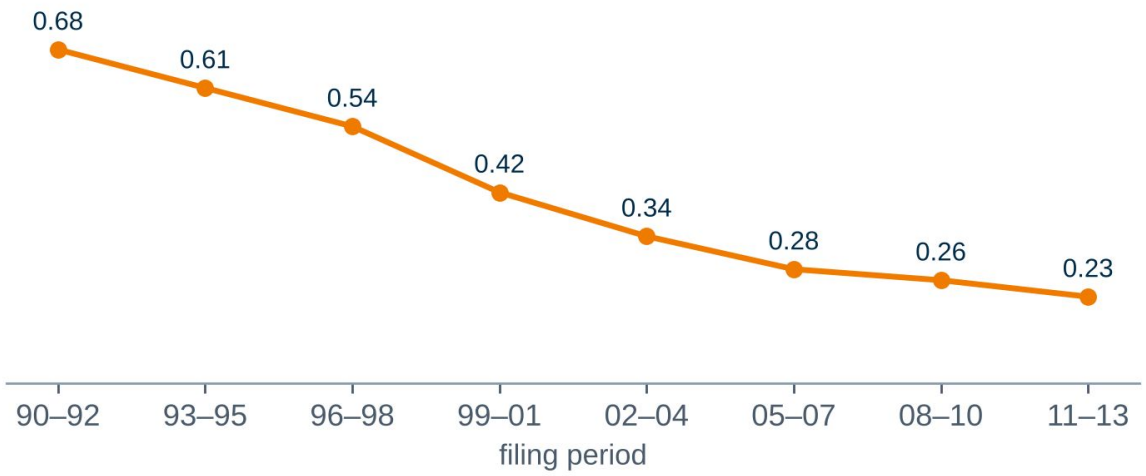
OECD indicator · Forward citations: citations received by EPO patents and their equivalents within **5** or **7 years of publication**.

Dimension **Technological**

PATSTAT Data Catalog: TLS212_CITATION · TLS201_APPLN · TLS209_APPLN_IPC

Comparing forward citations fairly

Average forward citations: younger patents had less time



Two problems

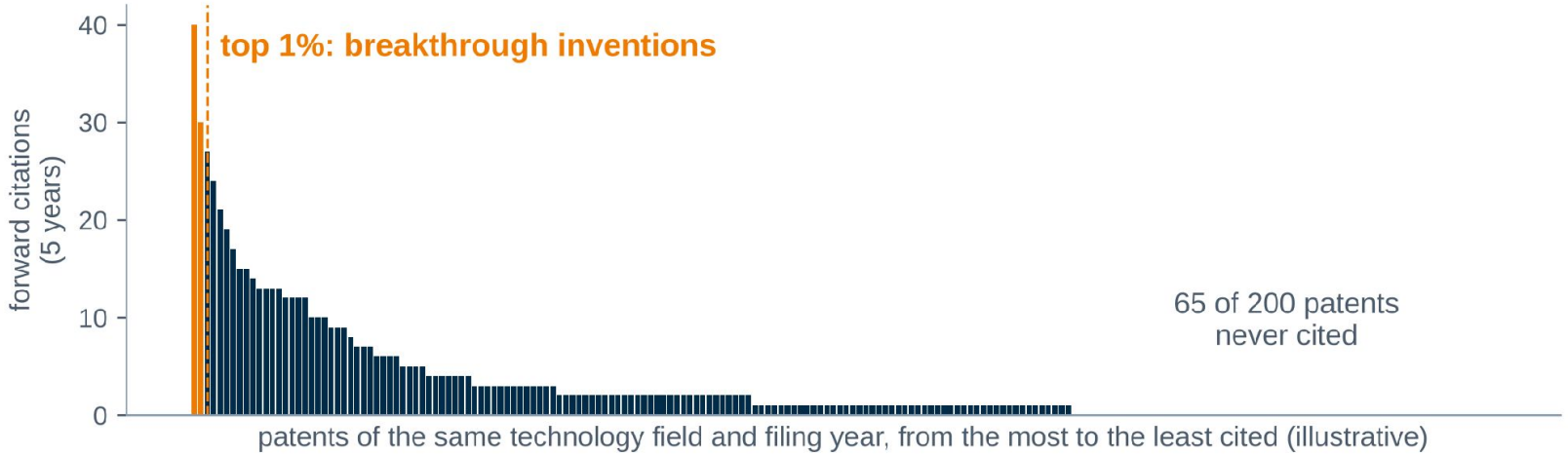
Truncation: we only see the citations received so far; younger patents had less time.

Fields differ: some fields cite much more than others, and practices change over time.

The fix: count within a fixed window (e.g. 5 years), then divide by the average of patents of the same field and year.

	Citations in 5 years	Field-year average	Normalised
Patent A · telecom 2010	4	2.0	2.0
Patent B · pharma 2015	4	0.8	5.0

Breakthrough inventions: the top 1%



What it measures: whether the patent is among the 1% most cited patents of its technology field and filing year.

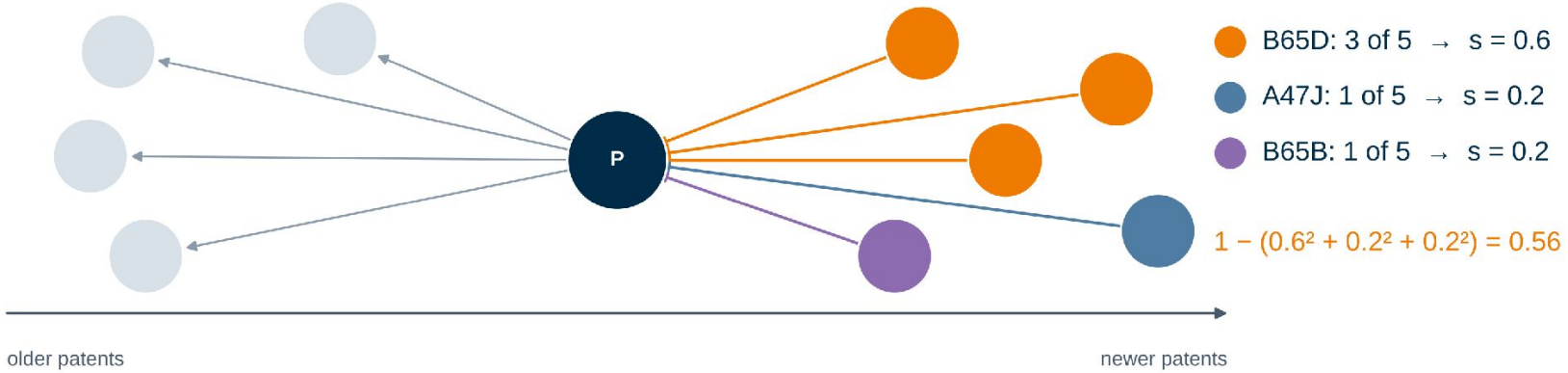
Why it signals value: the distribution of value is extremely skewed: a few patents carry most of the value, and the most cited are the ones later inventions are built on.

How to measure it: information available in OECD Patent Quality Dataset

OECD **OECD indicator · Breakthrough inventions:** 1 if the patent is in the top 1% most cited of its cohort (5-year window);

Squicciarini, Dernis & Criscuolo (2013), OECD STI Working Paper 2013/03, p. 42; Ahuja & Lampert (2001), Strategic Management Journal.

Generality: how widely is it used?



$$G_X = 1 - \sum_{j=1}^{M} \left(\frac{1}{N} \sum_{i=1}^{N} \beta_{ji} \right)^2, \qquad \beta_{ji} = \frac{T_{ji}^n}{T_i^n}$$

N patents cite X within 5 years; T_i^n: IPC classes (most detailed level) of citing patent i

T_{ji}^n: those inside the 4-digit class j; M: number of 4-digit classes

What it measures: how spread over different technology fields the patents that cite p are.

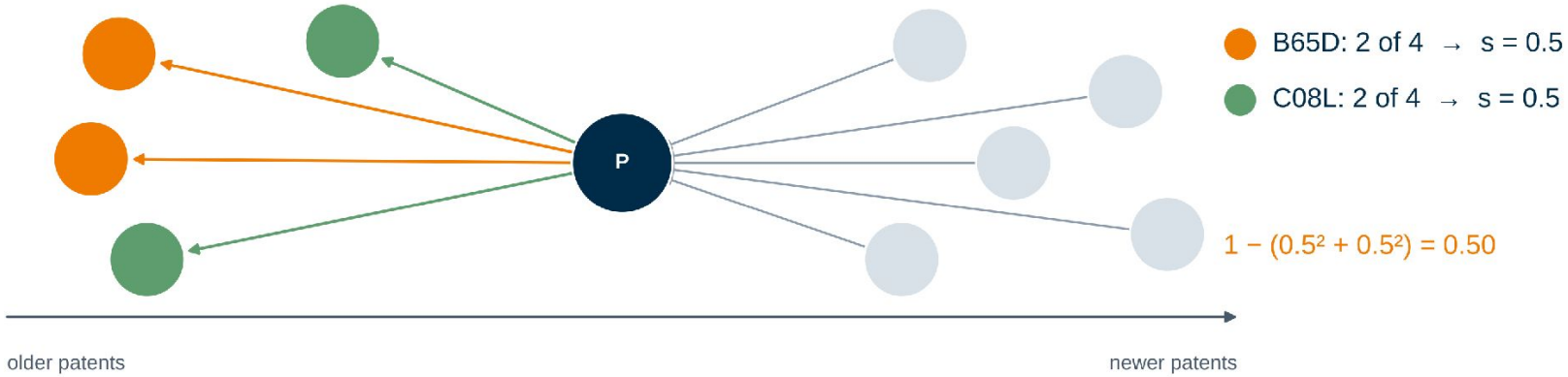
Why it signals value: a patent used in many fields has many potential applications and markets, like a general-purpose technology.

How to measure it: available in OECD Patent Quality Dataset

Watch out: it depends on how fine the classes are, and **treats close and distant fields alike**; with few citations it is unstable.

OECD indicator · Generality index: dispersion of the 4-digit IPC classes of the citing patents, weighted by all their classes; 5-year window

Originality: combining distant knowledge



$$\text{Originality}_p = 1 - \sum_{j=1}^{n_p} s_{pj}^2$$

s_{pj}: share of the patents cited by p that belong to IPC class j

n_p: number of classes of the cited patents; 0 = one field, close to 1 = many fields

What it measures: how spread over different technology fields the patents cited by p are.

Why it signals value: an invention that recombines knowledge from distant fields is more likely to be new and important.

How to measure it: available in OECD Patent Quality Dataset.

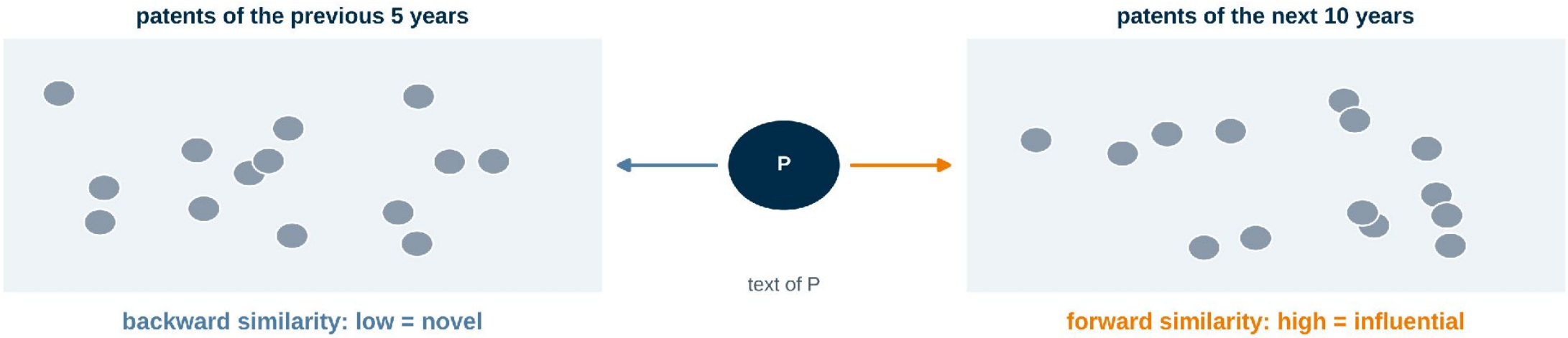
Watch out: few backward citations make it unstable (same correction as generality); examiners add many of the citations.

OECD **OECD indicator · Originality index:** dispersion of the 4-digit (or 7-digit) IPC classes of the cited patents

Dimension **Technological**

PATSTAT Data Catalog: TLS202_APPLN_TITLE · TLS203_APPLN_ABSTR

Text novelty: how new are the words?



What it measures: how different the text of P is from earlier patents (novelty), and how much later patents resemble it (impact).

Why it signals value: a patent that uses new words and concepts, later taken up by others, opened a new direction.

How to measure it: embeddings of title and abstract (TLS202, TLS203) as in Tutorial 2; cosine similarity with the patents of the 5 years before and the 10 after.

Watch out: new words are not always new ideas; results depend on the text model; the forward part needs years.

05

Legal value indicators

Grant, examination, opposition, claims and their text

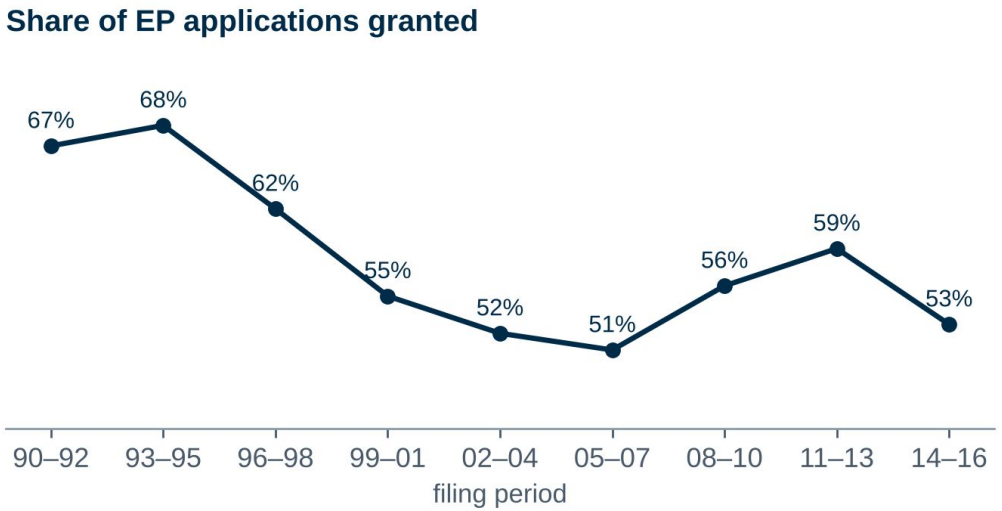


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Dimension **Legal**

PATSTAT Data Catalog: TLS201_APPLN · TLS211_PAT_PUBLN

Grant status



What it measures: whether the application, or a member of its family, has been granted.

Why it signals value: the invention passed the tests of novelty, inventive step and industrial application: the right can be enforced.

How to measure it: granted in TLS201, or a publication with publn_first_grant = 'Y' in TLS211; for the family, any member.

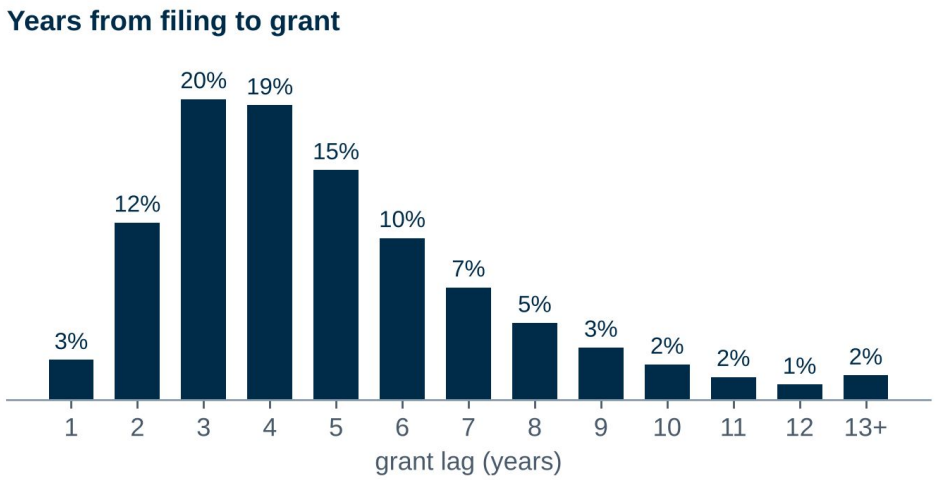
Watch out: recent applications are often still pending, so compare applications of the same year; a pending application can still be valuable.

EPO data, via A. De Marco.

Dimension **Legal**

PATSTAT Data Catalog: TLS201_APPLN · TLS211_PAT_PUBLN

Grant lag and examination



What it measures: how long and how hard the examination was: years from filing to grant, number of exchanges with the examiner.

Why it signals value: a hard-fought examination means the claims were tested against the prior art; a long lag also signals complex, uncertain technologies.

How to measure it: grant date minus filing date (TLS211 and TLS201); the exchanges with the examiner are in the EP Register and in the USPTO examination data.

Watch out: applicants can slow the examination on purpose, to keep the claims open while the market evolves.

OECD indicator · Grant lag index: 1 − days from filing to grant / the longest delay in the cohort: high when granted fast, as important patents are

Dimension **Legal**

PATSTAT Data Catalog: TLS231_INPADOC_LEGAL_EVENT

Opposition and litigation

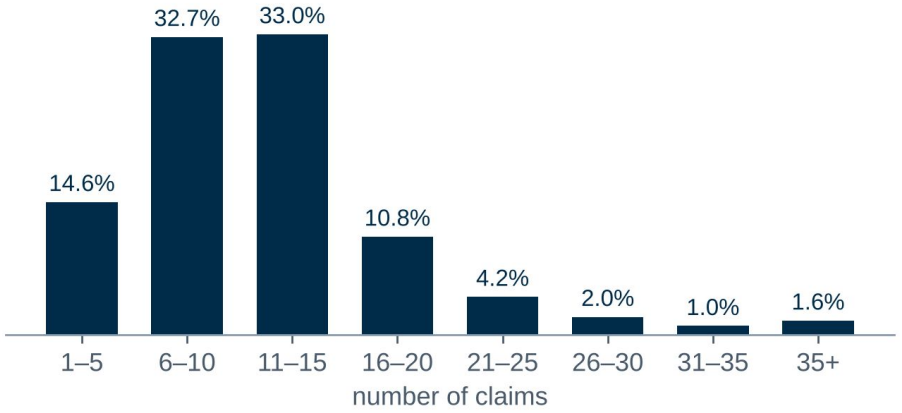
What it measures	Whether rivals tried to revoke the patent (opposition at the EPO within nine months of the grant) or the patent went to court.
Why it signals value	Nobody pays to attack a patent that does not hurt them; a patent that survives an attack is a tested patent.
How to measure it	EP oppositions: legal events in TLS231 or the EP Register; litigation: court databases (e.g. *Darts-IP*, the *US PACER*).

Dimension **Legal** ·

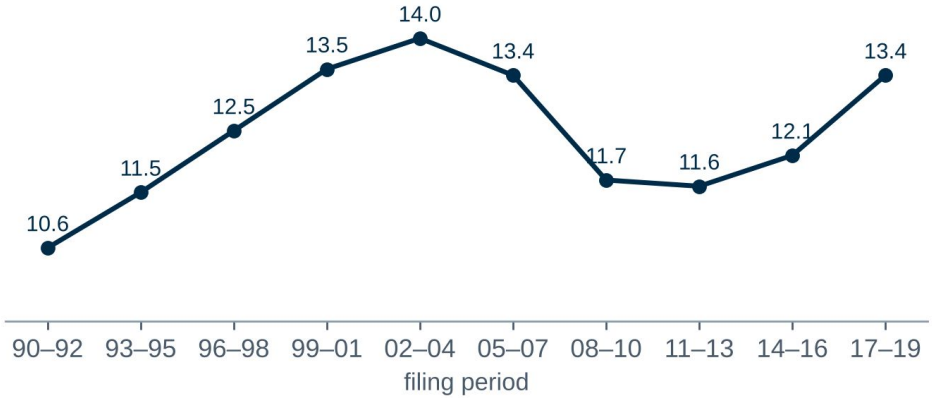
PATSTAT Data Catalog: TLS211_PAT_PUBLN

Number of claims: the legal scope

Claims per EP patent



Average number of claims



What it measures: how many claims the patent has: the claims draw the boundary of what is protected.

Why it signals value: more claims, more ways to stop imitators, more power on the market, and a higher chance of litigation.

How to measure it: publn_claims in TLS211 (the granted publication); count the independent claims too, from the full text.

Watch out: some applicants inflate claims for strategic reasons.

OECD indicator · Claims: number of claims

Claim text: the length of the first claim

Broad: 16 words

1. A capsule for preparing a beverage, comprising a cup-shaped body, a beverage ingredient and a lid.

Narrow: 52 words

1. A capsule for preparing a beverage, comprising a cup-shaped body made of aluminium, a ground coffee ingredient, a lid sealed to a flange of the body, and a filter placed between the ingredient and the lid, wherein the lid is pierced by pressure of the water at between 8 and 12 bar.

What it measures: the number of words of the first independent claim, the broadest one.

Why it signals value: each word is a limitation: a short claim covers more products, so it blocks more rivals; a long one is easy to design around.

How to measure it: from the full text of the claims (EP full text, Tutorial 2): count the words of claim 1.

Watch out: compare claims in the same language and field

(Extra) Claim text: readability (1/2), what SMOG and CLI measure

SMOG: how many long words?

$$\text{SMOG} = 1.043\sqrt{\text{polysyllables} \times \frac{30}{\text{sentences}}} + 3.1291$$

Count the words with three or more syllables (be-ve-rage, a-lu-mi-ni-um), scale the count to 30 sentences, take the square root.

→ more long words per sentence → higher score

→ a score of 16 means: you need a university degree to read it comfortably

Paper: Identification of valuable patents: New indicators and effects of prediction time points (Lai et al., 2026)

CLI: how long are words and sentences?

$$\text{CLI} = 0.0588\,L - 0.296\,S - 15.8$$

L = average number of letters per 100 words; S = average number of sentences per 100 words.

→ longer words (more letters) → higher score

→ more sentences for the same words, i.e. shorter sentences → lower score

06

Economic value indicators

Renewals, family, market reaction, transfers, products

Politecnico di Torino

Dimension **Economic** · **Legal**

(PATSTAT Data Catalog: TLS231_INPADOC_LEGAL_EVEN)T

Renewals: owners vote with their money

What it measures	How long the owner keeps paying the renewal fees, and how many years of protection are left.
Why it signals value	Fees grow every year; an owner pays only while the patent is worth more than the fee.
How to measure it	Lapses for non-payment are legal events (TLS231, INPADOC legal status); residual life = 20 years from filing minus age, if alive.
Watch out	Renewals are observed only after the fact

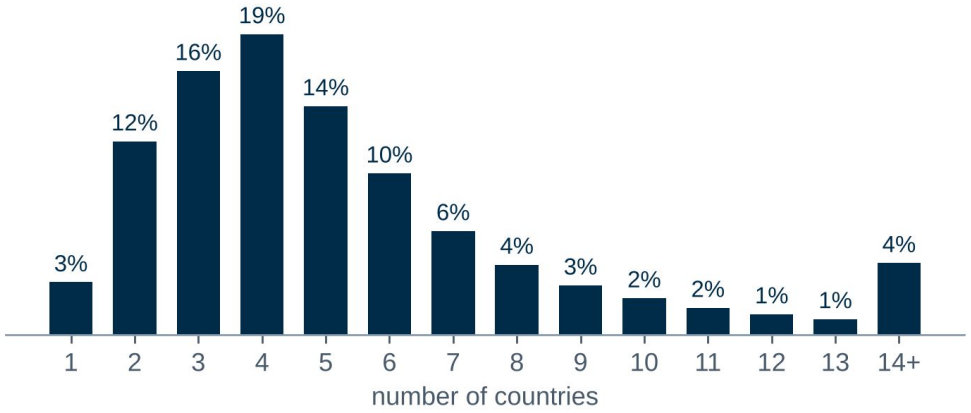
OECD indicator · Patent renewal: number of years a granted EP patent has been kept alive, counted from the application year.

Dimension **Economic I**

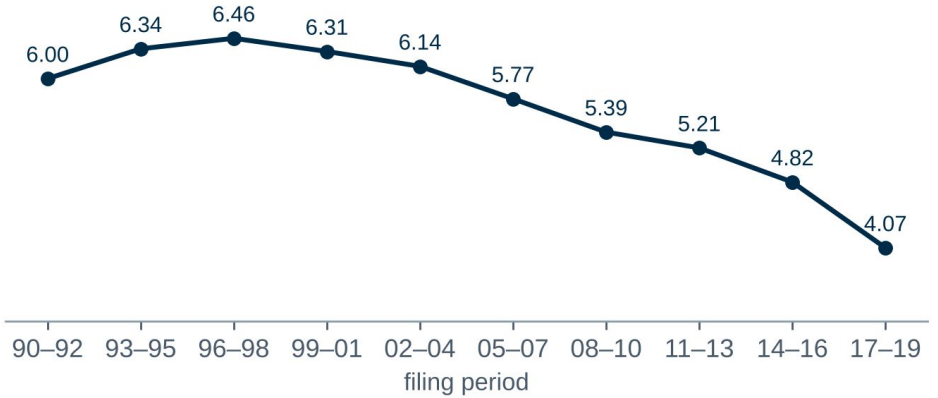
PATSTAT Data Catalog: TLS201_APPLN

Family size: how many countries?

Countries in the patent family



Average number of countries per family



What it measures: in how many countries, or at how many offices, the same invention is protected.

Why it signals value: filing abroad is expensive: owners extend only the inventions they expect to be valuable, and the size of the protected market grows.

How to measure it: docdb_family_size in TLS201, or count the distinct appln_auth of the family (Tutorial 1).

Watch out: say which family you count (DOCDB, INPADOC, countries): the numbers differ. Recent families are still growing.

OECD indicator · Family size: number of patent offices at which the invention is protected.

PATSTAT Data Catalog: TLS201_APPLN · TLS207_PERS_APPLN · TLS206_PERSON

Co-applicants: collaboration

What it measures	How many firms or institutions own the patent together.
Why it signals value	Several owners pool money and know-how to develop and exploit the invention.
How to measure it	nb_applicants in TLS201, or TLS207 with applt_seq_nr > 0; harmonised names in TLS206 (han_name/psn_name).
Watch out	94% of EP patents have one applicant: partners often leave the ownership to one of them, so collaboration is underestimated. Shared ownership can also make licensing harder.

Market reaction to the grant

What it measures	How much the stock-market value of the owner changes when the patent is granted.
Why it signals value	Investors put money on what they expect the patent to earn: a direct estimate in dollars.
How to measure it	Excess stock return in the days around the grant × market value of the firm (data of Kogan et al. for US listed firms).
Watch out	Only listed firms; the grant is often expected, so part of the value is already in the price.

Dimension **Economic** · **Legal**

PATSTAT Data Catalog: TLS231_INPADOC_LEGAL_EVENT

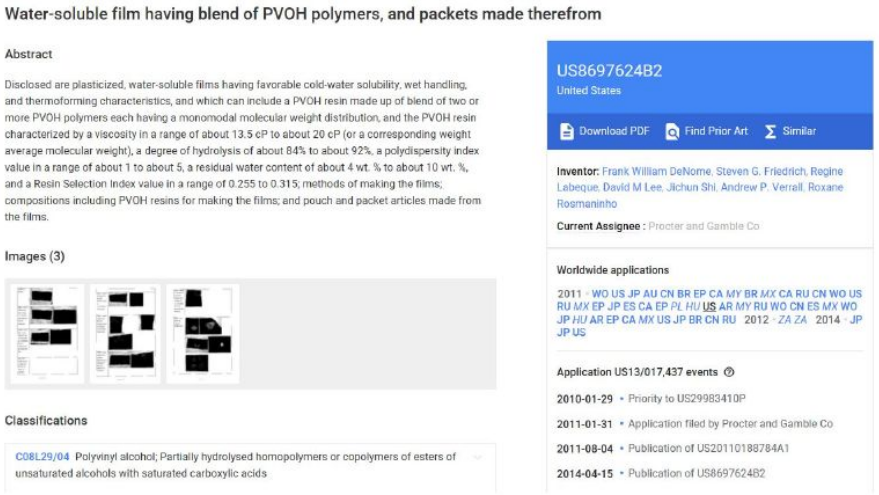
Transfers and licences

What it measures	Whether the patent changed owner, or was licensed.
Why it signals value	Someone paid to own it or use it: a revealed market value.
How to measure it	Changes of owner: legal events in TLS231, the EP Register, the USPTO Patent Assignment Dataset (PAD); licences: patent registers, annual reports of listed firms.
Watch out	Prices are almost never public; transfers inside a group of firms are not sales.

Dimension **Economic**

PATSTAT Data Catalog: TLS202_APPLN_TITLE · TLS203_APPLN_ABSTR

Products on the market: from patents to products



Water-soluble film having blend of PVOH polymers, and packets made therefrom

Abstract

Disclosed are plasticized, water-soluble films having favorable cold-water solubility, wet handling, and thermoforming characteristics, and which can include a PVOH resin made up of blend of two or more PVOH polymers each having a monomodal molecular weight distribution, and the PVOH resin characterized by a viscosity in a range of about 13.5 cP to about 20 cP (or a corresponding weight average molecular weight), a degree of hydrolysis of about 84% to about 92%, a polydispersity index value in a range of about 1 to about 5, a residual water content of about 4 wt. % to about 10 wt. %, and a Resin Selection Index value in a range of 0.255 to 0.315; methods of making the films; compositions including PVOH resins for making the films; and pouch and packet articles made from the films.

Images (3)

Classifications

C08L29/04 Polyvinyl alcohol; Partially hydrolysed homopolymers or copolymers of esters of unsaturated alcohols with saturated carboxylic acids

US8697624B2
United States

Download PDF · Find Prior Art · Similar

Inventor: Frank William DeNome, Steven G. Friedrich, Regine Labeque, David M Lee, Jichun Shi, Andrew P. Verrall, Roxane Rosmaninho

Current Assignee : Procter and Gamble Co

Worldwide applications

2011 · WO US JP AU CN BR EP CA MY BR MX CA RU CN WO US RU MX EP JP ES CA EP PL HU US AR MY RU WO CN ES MX WO JP HU AR EP CA MX US JP BR CN RU · 2012 · ZA ZA · 2014 · JP JP US

Application US13/017,437 events

2010-01-29 • Priority to US29983410P

2011-01-31 • Application filed by Procter and Gamble Co

2011-08-04 • Publication of US20110188784A1

2014-04-15 • Publication of US8697624B2

(a) Patent application in 2011



(b) The first Tide Pods in 2012

54%

of new products come from firms that never patent (65% of quality-adjusted innovation).

+0.4%

new products the next year for every 10% more patent filings by a firm in a category.

6×

weaker link for the largest firms than for the smallest: leaders' patents often do not become products.

Strategic

leaders' patents raise revenues even without new products, and are followed by fewer products from rivals.

For valuation: a patent linked to a product protects real sales; one that is not may still block rivals.

Where to find each indicator: technological

Indicator	From PATSTAT	Ready-made
Inventors	TLS201: nb_inventors; TLS207 with invt_seq_nr > 0	—
Technological scope	TLS224 (CPC) or TLS209 (IPC): distinct subclasses	OECD PQI: patent scope
Backward citations	TLS212 → TLS211 → TLS201 (Tutorial 3)	OECD PQI: backward citations; OECD Citations DB
Citations to science	TLS212 with cited_npl_publn_id ≠ '0' → TLS214	OECD PQI: NPL citations; Reliance on Science (Marx & Fuegi)
Forward citations	TLS212; TLS201: nb_citing_docdb_fam	OECD PQI: forward citations (5 and 7 years); OECD Citations DB
Breakthrough inventions	forward citations ranked by field and year	OECD PQI: breakthrough (top 1%)
Generality, originality	TLS212 + TLS209 (IPC of citing / cited patents)	OECD PQI: generality, originality, radicalness
Text novelty	TLS202, TLS203: titles and abstracts (Tutorial 2)	

PATSTAT Data Catalog: TLS201_APPLN · TLS202_APPLN_TITLE · TLS203_APPLN_ABSTR · TLS207_PERS_APPLN · TLS209_APPLN_IPC · TLS211_PAT_PUBLN · TLS212_CITATION · TLS214_NPL_PUBLN · TLS224_APPLN_CPC

Where to find each indicator: legal and economic

Indicator	From PATSTAT	Ready-made, other sources
Grant status	TLS201: granted; TLS211: publn_first_grant	—
Grant lag, examination	TLS201 filing date, TLS211 grant date	OECD PQI: grant lag; EP Register; USPTO PatEx
Opposition, litigation	TLS231: legal events (oppositions)	EP Register; Darts-IP; PACER (US courts)
Number of claims	TLS211: publn_claims	OECD PQI: claims
Claim text: length, readability, abstractness	— (full text: EPAB (TIP), Tutorial 2)	USPTO Patent Claims Research Dataset
Renewals	TLS231: INPADOC legal status (fees, lapses)	OECD PQI: renewal
Family size	TLS201: docdb_family_size; appln_auth of the family	OECD PQI: family size
Co-applicants	TLS201: nb_applicants; TLS207, TLS206 (han_name/psn_name)	—
Market reaction to the grant	—	—-
Transfers and licences	TLS231: changes of owner	USPTO Patent Assignment Dataset; EP Register
Products on the market	TLS202, TLS203 + product descriptions	IPRoduct (EPFL)- not free

PATSTAT Data Catalog: TLS201_APPLN · TLS202_APPLN_TITLE · TLS203_APPLN_ABSTR · TLS206_PERSON · TLS207_PERS_APPLN · TLS211_PAT_PUBLN · TLS231_INPADOC_LEGAL_EVENT

OECD Patent Quality Indicators; EPO, European Patent Register; USPTO research datasets (PatEx, Patent Claims, Patent Assignment); Kogan et al. (2017); Argente et al. (2025).

07

Scoring patents

Rule-based algorithms, tools, the IP scoreboard



Politecnico
di Torino

Rule-based algorithms: combining the indicators

Step 1 · Make the indicators comparable: normalize among year and fields

Step 2 · Combine them: add them up, with or without weights, or apply rules.

1 · Without weights

Every indicator counts the same: w = 1/K.

OECD composite quality index: forward citations, family size, claims, generality (+ backward citations and grant lag), normalised by field and year, equal importance.

2 · With weights

Experts decide how much each indicator counts.

IP scoreboard of Politecnico di Torino and Leonardo (high, medium, low); EPO IP Score; the Italian valuation grid.

3 · Rules and thresholds

Conditions that select or flag patents.

'Granted and family of 4 or more'; 'top 10% of citations in its field'; 'opposed and maintained'.

Squicciarini, Dernis & Criscuolo (2013), OECD STI Working Paper 2013/03, pp. 59–63: patent quality composite index.

Ready-made tools to rate patents

Tool	Provider	What it rates
IP Score	European Patent Office	patents, technologies, research projects
IP Health Check	UK Intellectual Property Office	patents, trademarks, designs, copyright
Valuation grid	MIMIT, UIBM, with ABI, CRUI, CGII	patents and technologies
IP scoreboard	Politecnico di Torino with Leonardo	patents

Links: EPO IP Score · UK IPO IP Health Check ·

The IP scoreboard: three groups of indicators

Legal robustness and coverage

legal dimension · public patent data

Grant status	H
Legal strength (opposition, litigation)	H
Family size	M
Residual life	L
Search report quality	L
Research effort (inventors)	L
Collaboration (co-applicants)	L

Technological relevance

technological dimension · public patent data

Technical merit (citations received)	H
Technological scope (classes)	M
Legal scope (claims)	M
Innovativeness (patents cited)	L
Science basicness (papers cited)	L

Strategic relevance

economic dimension · private owner and market data

Value of the product business plan	H
R&D intensity	H
Growth of the field	H
Impact on capabilities	M
Competition	M
Concentration	L
Licensing	L

Weight: **H** high · **M** medium · **L** low

IP scoreboard, Politecnico di Torino and Leonardo; via A. De Marco.

A worked example: two patents, one scoreboard

Indicator	Weight	Patent A: data	Score	Patent B: data	Score
Grant status	3	granted	5	granted	5
Legal strength	3	opposition survived	2	never opposed	0
Technical merit	3	3 citations received	2	11 citations received	5
Family size	2	6 members	3	3 members	2
Legal scope	2	18 claims	5	9 claims	0
Technological scope	2	2 subclasses	0	5 subclasses	5
Research effort · collaboration	1 · 1	5 inventors · 2 applicants	3 · 5	1 inventor · 1 applicant	0 · 0
Innovativeness · science	1 · 1	6 patents · 0 papers cited	3 · 0	2 patents · 4 papers cited	1 · 4
Residual life	1	12 years	3	16 years	5
Weighted total (out of 100)			**57**		**54**

Same total, opposite profiles: A is legally robust, B is technologically important. Is one number enough?

Questions?

Next PW tutorial
Patent valuation: designing an algorithm · Monday 9 November, 13:00–14:30 · Room 4P
Then
PW Tutorial of patent valuation (II) · Monday 23 November, 13:00–14:30 · Room 4P



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